

Circuit Simulation Software

Simulating a circuit's behaviour before actually building it greatly improves efficiency and provides insights into the behaviour of electronic circuit designs

■ NIRAJ SAHAY

The best method to check a design in electronics is to use a circuit simulator and make a simple model of it to see how it works before assembling the real circuit. Electronic circuit simulation tools use mathematical models to replicate the behaviour of an actual electronic device or circuit. Simulating a circuit's behaviour before actually building it greatly improves efficiency and provides insights into the behaviour of electronic circuit designs.

Therefore, in this month's EFY-CD, we have focused on electronic circuit simulation software, some of which are available for free over the Internet.

Electric 8.11



This open-source electronic design automation (EDA) system can handle many forms of circuit design and simulation.

What's Special? The Electric VLSI design system is highly flexible and powerful. It comes with features like custom IC layout, schematic capture for digital and analogue circuits, and textual languages such as VHDL and Verilog. Layout is done by placing and wiring electrical components. Electric comes with a built-in 12-state switch-level simulator, called ALS. The simulator displays waveforms in a separate window and lets users cross-probe from either the waveform or the circuit window.

Home Page: www.staticfreesoft.com/
Download URL: www.staticfreesoft.com/products/Free.html
Platforms Supported: Linux, Mac, MS Windows

Logisim



Logisim is a graphical application for designing and simulating digital logic circuits.

What's Special? With Logisim you can build larger circuits from smaller sub-circuits, and draw bundles of wires with a single mouse drag. Logisim can be used to design and simulate entire CPUs for educational purposes. Its simple interface with toolbar and ability to simulate circuits as you build them, make it an application that is simple enough to learn and use.

Home Page: <http://ozark.hendrix.edu/~burch/logisim/>
Download URL: <http://ozark.hendrix.edu/~burch/logisim/download.html>
Platforms Supported: Linux, Mac, MS Windows. Logisim requires Java 1.4 or later

Ngspice



Based on the Berkeley's Spice3f5, Ngspice is a mixed-mode, mixed-level circuit simulator. It is being developed to include new features to existing Spice3f5 and also to fix its bugs.

What's Special? It is a part of the popular open source gEDA project. Its code is based on three open source software packages: Spice3f5, Cider1b1 and Xspice.

Ngspice has built-in models for semiconductor devices, and the user needs to specify only the pertinent model parameter values. It provides a direct link between technology parameters and circuit performance. A mixed-level circuit and device simulator can provide greater simulation accuracy than a standalone circuit or device simulator by numerically modeling the critical devices in a circuit.

Home Page: <http://ngspice.sourceforge.net/>

Download URL: <http://ngspice.sourceforge.net/download.html>

Platforms Supported: LINUX, MS Windows, MAC and BSD

Simulo v2.5



Simulo digital circuit simulator developed by Benjamin Gentner can simulate circuit operation.

What's Special? It's easy to use and is designed for students and hobbyists. It includes many digital items such as seven-segment display and 4-bit shift register. The diagram created can be exported to an image and the application also supports variable clock speed along with labeling of inputs/outputs.

Home Page: <http://simulo.codeplex.com/>
Download URL: <http://simulo.codeplex.com/releases/view/46430>

Platforms Supported: Windows XP (SP2 or later), Windows Vista, Windows Server (2003 SP1 or later), Windows 7. Requires .NET Framework 3.5 SP1 or newer

Qucs



Quite universal circuit simulator (Qucs) is an open source software.

What's Special? The software aims to support all kinds of circuit simulation types, e.g. DC, AC, S-parameter, transient, noise and harmonic balance analysis. Pure digital simulations are also supported using VHDL and/or Verilog.

Home Page: <http://qucs.sourceforge.net/>
Download URL: <http://qucs.sourceforge.net/download.html>

Platforms Supported: MS Windows, MAC OS and different Linux-based OS

The author is an electronics enthusiast working as system administrator at EFY